MILITARY SPECIFICATION SHEET

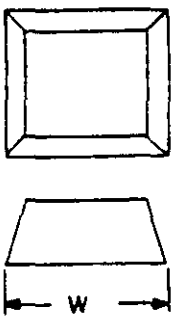
CAPACITORS, CHIP, SINGLE LAYER, FIXED UNENCAPSULATED, CERAMIC DIELECTRIC, ESTABLISHED RELIABILITY, STYLE CPCR01 (HIGH FREQUENCY)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

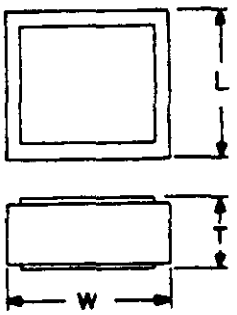
The requirement for acquiring the capacitors described herein shall consist of this specification and the latest issue of MIL-C-49464.

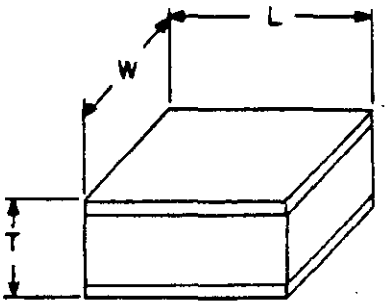
OPTIONAL CONFIGURATION, SEE NOTE 3.

TRAPEZOID



STRAIGHT SIDED BORDERED





Dimensions			
Case code	W	L (max)	T
A	.010 +0, -.003	.010 ±.005	.004 ±.001
B	.015 +0, -.003	.020 ±.010	.006 ±.002
C	.020 +0, -.005	.020 ±.010	.006 ±.002
D	.025 +0, -.003	.030 ±.015	.006 ±.002
E	.030 +0, -.003	.030 ±.015	.006 ±.002
F	.035 ±.005	.040 ±.020	.007 ±.003
G	.050 ±.010	.060 ±.030	.007 ±.003
H	.070 ±.010	.080 ±.040	.008 ±.003
J	.090 ±.010	.100 ±.050	.010 ±.004

Inches	mm
.001	0.03
.002	0.05
.003	0.08
.004	0.10
.005	0.13
.006	0.15
.007	0.18
.008	0.20
.010	0.25
.015	0.38
.020	0.51
.025	0.64
.030	0.76
.035	0.89
.040	1.02
.050	1.27
.060	1.52
.070	1.78
.080	2.03
.090	2.29
.100	2.54

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Alternate configurations are allowable if they fall within the envelope given and the performance characteristics are not affected.

FIGURE 1. CPCR01 capacitors.

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REQUIREMENTS:

Dimensions and configuration: See figure 1.

Case type: Single layer, unencapsulated.

Material: Ceramic.

Termination: N, 50 microinches gold over a suitable barrier base metal.

Capacitance: See table I.

Rated voltage: See table I.

Capacitance tolerance: See table II.

Operating temperature range: -55°C to +125°C.

Temperature coefficient, case code, minimum and maximum capacitance: See tables III and IV.

TABLE I. Style CPCR01 electrical characteristics and dash numbers.

Dash number				Cap.	Volt
M	P	R	S	pF	
001	201	401	601	0.05	50
002	202	402	602	0.06	50
003	203	403	603	0.07	50
004	204	404	604	0.08	50
005	205	405	605	0.09	50
006	206	406	606	0.1	50
007	207	407	607	0.2	50
008	208	408	608	0.3	50
009	209	409	609	0.4	50
010	210	410	610	0.5	50
011	211	411	611	0.6	50
012	212	412	612	0.7	50
013	213	413	613	0.8	50
014	214	414	614	0.9	50
015	215	415	615	1.0	50
016	216	416	616	1.1	50
017	217	417	617	1.2	50
018	218	418	618	1.3	50
019	219	419	619	1.4	50
020	220	420	620	1.5	50
021	221	421	621	1.6	50
022	222	422	622	1.7	50
023	223	423	623	1.8	50
024	224	424	624	1.9	50
025	225	425	625	2.0	50
026	226	426	626	2.1	50
027	227	427	627	2.2	50
028	228	428	628	2.4	50
029	229	429	629	2.7	50
030	230	430	630	3.0	50
031	231	431	631	3.3	50
032	232	432	632	3.6	50
033	233	433	633	3.9	50
034	234	434	634	4.3	50
035	235	435	635	4.7	50
036	236	436	636	5.1	50
037	237	437	637	5.6	50
038	238	438	638	6.2	50
039	239	439	639	6.8	50
040	240	440	640	7.5	50

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TABLE I. Style CPCR01 electrical characteristics and dash numbers - Continued.

Dash number				Cap.	Volt
M	P	R	S	pF	
041	241	441	641	8.2	50
042	242	442	642	9.1	50
043	243	443	643	10	50
044	244	444	644	11	50
045	245	445	645	12	50
046	246	446	646	13	50
047	247	447	647	15	50
048	248	448	648	16	50
049	249	449	649	18	50
050	250	450	650	20	50
051	251	451	651	22	50
052	252	452	652	24	50
053	253	453	653	27	50
054	254	454	654	30	50
055	255	455	655	33	50
056	256	456	656	36	50
057	257	457	657	39	50
058	258	458	658	43	50
059	259	459	659	47	50
060	260	460	660	51	50
061	261	461	661	56	50
062	262	462	662	62	50
063	263	463	663	68	50
064	264	464	664	82	50
065	265	465	665	100	50
066	266	466	666	120	50
067	267	467	667	150	50
068	268	468	668	180	50
069	269	469	669	220	50
070	270	470	670	270	50
071	271	471	671	330	50
072	272	472	672	470	50
073	273	473	673	0.05	100
074	274	474	674	0.06	100
075	275	475	675	0.07	100
076	276	476	676	0.08	100
077	277	477	677	0.09	100
078	278	478	678	0.1	100
079	279	479	679	0.2	100
080	280	480	680	0.3	100
081	281	481	681	0.4	100
082	282	482	682	0.5	100
083	283	483	683	0.6	100
084	284	484	684	0.7	100
085	285	485	685	0.8	100
086	286	486	686	0.9	100
087	287	487	687	1.0	100
088	288	488	688	1.1	100
089	289	489	689	1.2	100
090	290	490	690	1.3	100
091	291	491	691	1.4	100
092	292	492	692	1.5	100
093	293	493	693	1.6	100
094	294	494	694	1.7	100
095	295	495	695	1.8	100
096	296	496	696	1.9	100
097	297	497	697	2.0	100
098	298	498	698	2.1	100
099	299	499	699	2.2	100
100	300	500	700	2.4	100

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TABLE I. Style CPCR01 electrical characteristics and dash numbers - Continued.

Dash number				Cap.	Volt
M	P	R	S	pF	
101	301	501	701	2.7	100
102	302	502	702	3.0	100
103	303	503	703	3.3	100
104	304	504	704	3.6	100
105	305	505	705	3.9	100
106	306	506	706	4.3	100
107	307	507	707	4.7	100
108	308	508	708	5.1	100
109	309	509	709	5.6	100
110	310	510	710	6.2	100
111	311	511	711	6.8	100
112	312	512	712	7.5	100
113	313	513	713	8.2	100
114	314	514	714	9.1	100
115	315	515	715	10	100
116	316	516	716	11	100
117	317	517	717	12	100
118	318	518	718	13	100
119	319	519	719	15	100
120	320	520	720	16	100
121	321	521	721	18	100
122	322	522	722	20	100
123	323	523	723	22	100
124	324	524	724	24	100
125	325	525	725	27	100
126	326	526	726	30	100
127	327	527	727	33	100
128	328	528	728	36	100
129	329	529	729	39	100
130	330	530	730	43	100
131	331	531	731	47	100
132	332	532	732	51	100
133	333	533	733	56	100
134	334	534	734	62	100
135	335	535	735	68	100
136	336	536	736	75	100
137	337	537	737	82	100
138	338	538	738	91	100
139	339	539	739	100	100
140	340	540	740	110	100
141	341	541	741	120	100
142	342	542	742	130	100
143	343	543	743	150	100
144	344	544	744	160	100
145	345	545	745	180	100
146	346	546	746	200	100
147	347	547	747	220	100
148	348	548	748	240	100
149	349	549	749	270	100
150	350	550	750	330	100
151	351	551	751	390	100
152	352	552	752	470	100
153	353	553	753	560	100
154	354	554	754	680	100
155	355	555	755	820	100
156	356	556	756	1000	100
157	357	557	757	1200	100
158	358	558	758	1500	100

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TABLE II. Capacitance tolerances.

Temperature coefficient symbol	Capacitance tolerances	
E, F, G, H	< 5 pF	P, A, B, C, D
	≥ 5 pF	F, G, J, K
J, K, L, M, N, P, R	< 10 pF	C, D
	≥ 10 pF	G, J, K, M
T, U, V, W	< 10 pF	C, D
	≥ 10 pF	J, K, M
X, Y		J, K, M
Z		M, V, Z

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TABLE III. Style CPCR01, temperature coefficient, case code, minimum and maximum capacitance 50 V dc.

Case code A			Case code B			Case code C		
Temperature coefficient	Minimum capacitance pF	Maximum capacitance pF	Temperature coefficient	Minimum capacitance pF	Maximum capacitance pF	Temperature coefficient	Minimum capacitance pF	Maximum capacitance pF
E	.05	.2	E	.07	.3	E	.1	.4
F	.05	.1	F	.08	.3	F	.1	.4
G	.09	.4	G	.2	.9	G	.2	1.2
H	.05	.2	H	.09	.5	H	.1	.7
J	.06	.3	J	.1	.6	J	.2	.8
K	.07	.3	K	.1	.7	K	.2	.9
L	.09	.4	L	.2	.8	L	.3	1.1
M	.1	.4	M	.2	1.0	M	.6	1.3
N	.3	.7	N	.5	1.5	N	.7	2.0
P	.3	1.0	P	.5	1.8	P	.7	3.0
R	.7	1.6	R	1.1	3.9	R	1.7	6.2
T	.8	2.7	T	1.7	5.6	T	2.7	7.5
U	1.0	5.6	U	2.1	10.0	U	3.3	12.0
V	1.6	3.3	V	2.2	8.2	V	2.7	10.0
W	1.0	6.8	W	1.4	18.0	W	2.7	22.0
X	2.4	15.0	X	4.7	33.0	X	8.2	47.0
Y	4.7	18.0	Y	6.2	47.0	Y	12.0	56.0
Z	12.0	56.0	Z	22.0	120.0	Z	33.0	180.0

Case code D			Case code E			Case code F		
Temperature coefficient	Minimum capacitance pF	Maximum capacitance pF	Temperature coefficient	Minimum capacitance pF	Maximum capacitance pF	Temperature coefficient	Minimum capacitance pF	Maximum capacitance pF
E	.2	.7	E	.2	.9	E	.3	1.5
F	.2	.7	F	.3	.8	F	.3	1.3
G	.8	2.4	G	.5	2.7	G	.4	4.3
H	.3	1.2	H	.4	1.5	H	.4	2.7
J	.3	1.5	J	.5	1.8	J	.5	3.3
K	.4	1.7	K	.5	2.1	K	.6	3.9
L	.5	2.1	L	.7	2.4	L	.7	4.7
M	.6	2.4	M	.8	3.0	M	.8	5.6
N	1.2	3.9	N	1.4	4.7	N	1.6	8.2
P	1.6	5.6	P	1.8	6.8	P	2.1	12.0
R	3.3	10.0	R	4.3	12.0	R	4.3	18.0
T	4.7	15.0	T	5.6	18.0	T	7.5	33.0
U	6.2	27.0	U	8.2	33.0	U	8.2	56.0
V	8.2	22.0	V	8.2	27.0	V	12.0	47.0
W	4.7	39.0	W	7.5	47.0	W	6.8	68.0
X	15.0	180.0	X	18.0	100.0	X	12.0	180.0
Y	22.0	100.0	Y	33.0	120.0	Y	33.0	220.0
Z	68.0	330.0	Z	82.0	330.0	Z	82.0	470.0

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TABLE III. Style CPCR01, temperature coefficient, case code, minimum and maximum capacitance 50 V dc - Continued.

Case code G			Case code H			Case code J		
Temperature coefficient	Minimum capacitance pF	Maximum capacitance pF	Temperature coefficient	Minimum capacitance pF	Maximum capacitance pF	Temperature coefficient	Minimum capacitance pF	Maximum capacitance pF
E	.5	3.3	E	.9	4.7	E	1.1	6.8
F	.7	2.7	F	1.2	4.7	F	1.4	5.6
G	.9	10.0	G	1.7	15.0	G	2.4	22.0
H	.7	5.6	H	1.4	8.2	H	1.8	12.0
J	1.0	8.2	J	1.8	11.0	J	2.4	15.0
K	1.1	8.2	K	2.0	12.0	K	2.7	18.0
L	1.4	10.0	L	2.4	15.0	L	3.3	22.0
M	1.6	12.0	M	3.0	18.0	M	3.6	27.0
N	3.3	18.0	N	6.2	27.0	N	8.2	39.0
P	4.3	27.0	P	9.1	39.0	P	12.0	56.0
R	9.1	47.0	R	16.0	68.0	R	20.0	82.0
T	15.0	68.0	T	24.0	100.0	T	33.0	120.0
U	18.0	120.0	U	30.0	150.0	U	39.0	270.0
V	18.0	82.0	V	56.0	120.0	V	68.0	180.0
W	12.0	150.0	W	24.0	270.0	W	33.0	470.0
X	27.0	330.0	X	47.0	470.0	X	62.0	470.0
Y	56.0	470.0	Y	120.0	470.0	Y	150.0	470.0
Z	150.0	470.0	Z	270.0	470.0	Z	270.0	470.0

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TABLE IV. Style CPCR01, temperature coefficient, case code, minimum and maximum capacitance 100 V dc.

Case code A			Case code B			Case code C		
Temperature coefficient	Minimum capacitance pF	Maximum capacitance pF	Temperature coefficient	Minimum capacitance pF	Maximum capacitance pF	Temperature coefficient	Minimum capacitance pF	Maximum capacitance pF
E	.05	.2	E	.07	.3	E	.1	.4
F	.05	.1	F	.08	.3	F	.1	.4
G	.09	.4	G	.2	.9	G	.2	1.2
H	.05	.2	H	.09	.5	H	.1	.7
J	.06	.3	J	.1	.6	J	.2	.8
K	.07	.3	K	.1	.7	K	.2	.9
L	.09	.4	L	.2	.8	L	.3	1.1
M	.1	.4	M	.2	1.0	M	.6	1.3
N	.3	.7	N	.5	1.5	N	.7	2.0
P	.3	1.0	P	.5	1.8	P	.7	3.0
R	.7	1.6	R	1.1	3.9	R	1.7	6.8
T	.8	2.7	T	1.7	5.6	T	2.7	8.2
U	1.0	5.1	U	2.1	10.0	U	3.3	12.0
V	1.6	3.3	V	2.2	8.2	V	2.7	10.0
W	1.0	6.8	W	1.4	16.0	W	2.7	22.0
X	2.4	15.0	X	4.7	33.0	X	8.2	47.0
Y	4.7	18.0	Y	6.2	47.0	Y	12.0	56.0
Z	12.0	56.0	Z	24.0	120.0	Z	33.0	180.0

Case code D			Case code E			Case code F		
Temperature coefficient	Minimum capacitance pF	Maximum capacitance pF	Temperature coefficient	Minimum capacitance pF	Maximum capacitance pF	Temperature coefficient	Minimum capacitance pF	Maximum capacitance pF
E	.2	.7	E	.2	.9	E	.3	1.5
F	.2	.7	F	.3	.8	F	.3	1.3
G	.8	2.4	G	.5	2.7	G	.4	4.3
H	.3	1.2	H	.4	1.5	H	.4	2.7
J	.3	1.5	J	.5	1.8	J	.5	3.3
K	.4	1.7	K	.5	2.1	K	.6	3.9
L	.5	2.1	L	.7	2.4	L	.7	4.7
M	.6	2.4	M	.8	3.0	M	.8	5.6
N	1.2	3.9	N	1.4	4.7	N	1.6	8.2
P	1.6	5.6	P	1.8	6.8	P	2.1	12.0
R	3.3	10.0	R	4.3	11.0	R	4.3	18.0
T	4.7	15.0	T	5.6	18.0	T	7.5	33.0
U	6.2	27.0	U	8.2	33.0	U	8.2	56.0
V	8.2	22.0	V	8.2	27.0	V	12.0	47.0
W	4.7	39.0	W	7.5	47.0	W	6.8	68.0
X	15.0	180.0	X	18.0	100.0	X	12.0	180.0
Y	20.0	100.0	Y	33.0	120.0	Y	30.0	220.0
Z	62.0	330.0	Z	75.0	390.0	Z	75.0	560.0

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TABLE III. Style CPCR01, temperature coefficient, case code, minimum and maximum capacitance 100 V dc - Continued.

Case code G			Case code H			Case code J		
Temperature coefficient	Minimum capacitance pF	Maximum capacitance pF	Temperature coefficient	Minimum capacitance pF	Maximum capacitance pF	Temperature coefficient	Minimum capacitance pF	Maximum capacitance pF
E	.5	3.3	E	.9	4.7	E	1.1	6.8
F	.7	2.7	F	1.2	4.3	F	1.4	5.6
G	.9	10.0	G	1.7	15.0	G	2.4	22.0
H	.7	5.6	H	1.4	8.2	H	1.8	12.0
J	1.0	8.2	J	1.8	11.0	J	2.4	15.0
K	1.1	8.2	K	2.0	12.0	K	2.7	18.0
L	1.4	10.0	L	2.4	15.0	L	3.3	22.0
M	1.6	12.0	M	3.0	18.0	M	3.6	27.0
N	3.3	18.0	N	6.2	27.0	N	8.2	36.0
P	4.3	27.0	P	9.1	39.0	P	12.0	56.0
R	9.1	47.0	R	15.0	68.0	R	20.0	82.0
T	15.0	68.0	T	24.0	100.0	T	33.0	120.0
U	18.0	110.0	U	30.0	160.0	U	39.0	240.0
V	18.0	91.0	V	56.0	130.0	V	68.0	180.0
W	12.0	150.0	W	24.0	270.0	W	36.0	470.0
X	24.0	390.0	X	47.0	560.0	X	62.0	820.0
Y	56.0	470.0	Y	110.0	820.0	Y	150.0	1000.0
Z	150.0	1200.0	Z	270.0	1500.0	Z	270.0	1500.0

CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 85
NASA - NA

Review activities:
Navy - AS
Air Force - 17, 99
DLA - ES

User activities:
Navy - MC, SH
Air Force - 11, 19

Preparing activity:
Army - ER

Agent:
DLA - ES

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